

Data Dictionary

This file explains the columns in `agent_tool_tasks.csv`.

The dataset has **3,491 rows** and **33 columns**. Each row is one agent task: a user query, the tools available to the agent, and the target label.

The dataset was extracted from the Berkeley Function Calling Leaderboard JSON files:

https://github.com/ShishirPatil/gorilla/tree/main/berkeley-function-call-leaderboard/bfcl_eval/data

You do not need the original JSON files for this assignment.

Important Terms

Term	Meaning
Query	The user request.
Tool	A function/API the agent could call.
Aspect	A broad topic detected by keyword rules, such as <code>math</code> , <code>finance</code> , <code>media</code> , or <code>travel_weather</code> .
Query/tool match	A measure of whether the query and the available tools seem to talk about similar topics.
Target	The value we want to predict. Here the target is <code>can_answer</code> .

Columns

Column	Type	Description
<code>task_uid</code>	string	Row ID. Do not use it as a model feature.
<code>is_live_benchmark</code>	0/1	1 if the task came from the live benchmark, 0 otherwise.

Column	Type	Description
task_complexity	category	low , medium , or high . This is a derived rough complexity bucket.
task_domain	category	Broad derived domain, such as math , travel , vision , nlp , or other .
query	text	Raw user request. Use it only to create features. Do not use the raw text directly in numeric models.
tool_names	text	Pipe-separated list of available tool names. Example: `math.factorial`
query_word_count	number	Number of words in the query.
num_available_tools	number	Number of tools available to the agent.
total_params	number	Total number of parameters across all available tools.
total_required_params	number	Total number of required parameters across all available tools.
avg_param_description_length	number	Average length of parameter descriptions. Has a few missing values.
tool_description_total_length	number	Total length of all tool descriptions.
query_aspects	category/text	Broad aspects detected in the query. Multiple aspects are separated by `
tool_aspects	category/text	Broad aspects detected in the tool names. Multiple aspects are separated by `
query_aspect_count	number	Number of aspects detected in the query.
tool_aspect_count	number	Number of aspects detected in the tool names.
aspect_overlap_count	number	Number of aspects that appear in both query_aspects and tool_aspects .
aspect_coverage_ratio	number	Fraction of query aspects that are also found in the tool aspects. Values are

Column	Type	Description
		between 0 and 1.
aspect_mismatch_count	number	Number of query aspects not found in the tool aspects.
query_tool_token_jaccard	number	Token-overlap score between the query and tool names. Higher means more shared words.
query_tool_action_overlap	number	Number of action words that appear in both the query and the tool names.
tool_action_verb_count	number	Number of action-like words in tool names, such as <code>get</code> , <code>search</code> , <code>send</code> , or <code>calculate</code> .
risky_tool_action_count	number	Number of tool-name action words that may involve risk, such as <code>send</code> , <code>book</code> , <code>buy</code> , <code>delete</code> , or <code>update</code> .
query_multi_intent_score	number	Count of words that suggest multiple steps, such as <code>also</code> , <code>then</code> , <code>first</code> , <code>next</code> , or <code>finally</code> .
query_specificity_score	number	Simple score based on numbers, quoted strings, and long tokens in the query.
query_unique_token_ratio	number	Number of unique query tokens divided by total query tokens.
query_code_signal	0/1	1 if the query contains code/API/data words such as <code>python</code> , <code>sql</code> , <code>json</code> , <code>api</code> , or <code>token</code> .
query_temporal_signal	0/1	1 if the query contains time or scheduling words such as <code>today</code> , <code>meeting</code> , <code>calendar</code> , or <code>deadline</code> .
query_sensitive_data_signal	0/1	1 if the query contains words such as <code>account</code> , <code>password</code> , <code>token</code> , <code>payment</code> , or <code>card</code> .
param_type_diversity	number	Number of different parameter type families used by the available tools.

Column	Type	Description
structured_param_ratio	number	Share of parameters that are arrays/lists or objects/dicts.
schema_rigidity_score	number	Rough score for how strict the tool schemas are. Higher means more required/limited parameters.
can_answer	0/1	Target label. 1 means at least one tool is relevant. 0 means the agent should refuse.

Notes About Feature Use

Column type	What to do
IDs	Do not use <code>task_uid</code> as a feature.
Raw text	Do not feed raw <code>query</code> or raw <code>tool_names</code> directly into numeric models. Use them only to create features.
Categorical columns	Encode columns such as <code>task_domain</code> , <code>task_complexity</code> , <code>query_aspects</code> , and <code>tool_aspects</code> before using them in models.
Multi-value columns	<code>query_aspects</code> , <code>tool_aspects</code> , and <code>tool_names</code> can contain several values separated by "pipe" (<code> </code>). Treat them carefully if you use them.
Missing values	Handle missing <code>avg_param_description_length</code> and missing/empty <code>tool_names</code> .
Target	Do not use <code>can_answer</code> as an input feature.

Target Meaning

can_answer	Meaning
1	At least one available tool is relevant to the user query.
0	No available tool is relevant; the agent should refuse to call a tool.